

MAT 1013: Discrete Mathematics

Session 4: Arguments and Valid Reasoning

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Learning Outcomes

By the end of this session, students should be able to:

1. Identify premises and conclusions in an argument
2. Translate simple arguments into symbolic form
3. Determine whether an argument is valid or invalid
4. Recognize common valid argument forms
5. Detect common invalid reasoning patterns
6. Explain validity using truth tables or logical structure

Part 1: What is an Argument?

Question

What is the difference between an argument in ordinary conversation and an argument in mathematics?

Note

In logic, an argument is a collection of statements where some statements are offered as reasons for another statement.

An **argument** is a sequence of propositions in which some propositions, called **premises**, are used to support another proposition, called the **conclusion**.

General Form

Premise 1
Premise 2
⋮
∴ Conclusion

The symbol $\therefore$ means “therefore.”

Example

If it rains, then the ground is wet.
It is raining.
 $\therefore$ The ground is wet.

Let

p : It rains, q : The ground is wet.

Then the argument becomes:

$p \rightarrow q$
 p
 $\therefore q$

Part 2: Validity

An argument is **valid** if whenever all the premises are true, the conclusion must also be true.

Important Point

Validity is about the **form** of the argument, not necessarily about whether the premises are actually true.

Example

All cats are birds.
Tom is a cat.
 $\therefore$ Tom is a bird.

This argument has false content, but the logical form is valid.

A valid argument can have false premises. But it cannot have true premises and a false conclusion.

Activity: Premise or Conclusion?

For each argument, identify the premises and the conclusion.

1. If a number is divisible by 4, then it is even. The number 12 is divisible by 4. Therefore, 12 is even.
2. All students in this class study mathematics. Nimal is a student in this class. Therefore, Nimal studies mathematics.
3. If I study, then I pass. I passed. Therefore, I studied.
4. If a function is differentiable, then it is continuous. This function is not continuous. Therefore, it is not differentiable.

Discussion :

Which of these arguments are valid? Justify.

Part 3: Common Valid Argument Forms

A. Propositional Argument Forms

1. Modus Ponens

$$\begin{array}{l} p \rightarrow q \\ p \\ \therefore q \end{array}$$

Example.

If n is divisible by 4, then n is even.
 n is divisible by 4.
 $\therefore n$ is even.

2. Modus Tollens

$$\begin{array}{l} p \rightarrow q \\ \neg q \\ \therefore \neg p \end{array}$$

Example.

If n is divisible by 4, then n is even.
 n is not even.
 $\therefore n$ is not divisible by 4.

3. Hypothetical Syllogism

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \therefore p \rightarrow r \end{array}$$

Example 1.

If n is divisible by 6, then n is divisible by 3.
If n is divisible by 3, then n^2 is divisible by 3.
 $\therefore$ If n is divisible by 6, then n^2 is divisible by 3.

Example 2.

If I study, then I pass.
If I pass, then I graduate.
 $\therefore$ If I study, then I graduate.

4. Disjunctive Syllogism

$$\begin{array}{l} p \vee q \\ \neg p \\ \therefore q \end{array}$$

Example.

n is even or n is odd.
 n is not even.
 $\therefore n$ is odd.

5. Addition

$$\begin{array}{l} p \\ \therefore p \vee q \end{array}$$

Example.

n is even.
 $\therefore n$ is even or n is prime.

6. Simplification

$$\begin{array}{l} p \wedge q \\ \therefore p \end{array}$$

Example.

n is even and $n > 10$.
 $\therefore n$ is even.

7. Conjunction

$$\begin{array}{l} p \\ q \\ \therefore p \wedge q \end{array}$$

Example.

n is even.
 $n > 10$.
 $\therefore n$ is even and $n > 10$.

8. Resolution

$$\begin{array}{l} p \vee q \\ \neg p \vee r \\ \therefore q \vee r \end{array}$$

Example.

n is even or n is prime.
 n is not even or $n > 1$.
 $\therefore n$ is prime or $n > 1$.

9. Constructive Dilemma

$$\begin{array}{l} p \rightarrow q \\ r \rightarrow s \\ p \vee r \\ \therefore q \vee s \end{array}$$

Example.

If n is divisible by 4, then n is even.
If n is prime and greater than 2, then n is odd.
 n is divisible by 4 or n is prime and greater than 2.
 $\therefore n$ is even or n is odd.

10. Destructive Dilemma

$$\begin{aligned}p &\rightarrow q \\r &\rightarrow s \\ \neg q &\vee \neg s \\ \therefore \neg p &\vee \neg r\end{aligned}$$

Example.

If n is divisible by 4, then n is even.

If n is prime and greater than 2, then n is odd.

n is not even or n is not odd.

$\therefore n$ is not divisible by 4 or n is not prime and greater than 2.

B. First-Order Logic Inference Rules

1. Universal Instantiation

$$\begin{aligned}\forall x P(x) \\ \therefore P(a)\end{aligned}$$

Example.

For every integer x , $x + 0 = x$.

$\therefore 5 + 0 = 5$.

2. Universal Generalization

$$\begin{aligned}P(x) \text{ has been proved for an arbitrary } x \\ \therefore \forall x P(x)\end{aligned}$$

Example.

Suppose x is an arbitrary real number. Since

$$x + 0 = x,$$

we conclude:

$$\forall x \in \mathbb{R}, x + 0 = x.$$

3. Existential Instantiation

$$\begin{aligned}\exists x P(x) \\ \therefore P(c)\end{aligned}$$

where c is a new symbol representing some object satisfying P .

Example.

There exists an integer x such that $x^2 = 4$.

Let c be such an integer.

$\therefore c^2 = 4$.

Note. We should not assume $c = 2$ unless this is given.

4. Existential Generalization

$$\begin{aligned}P(a) \\ \therefore \exists x P(x)\end{aligned}$$

Example.

$$2^2 = 4.$$

$$\therefore \exists x \in \mathbb{Z}, x^2 = 4.$$

C. Quantified Modus Ponens and Modus Tollens

1. Universal Modus Ponens

$$\begin{aligned} &\forall x (P(x) \rightarrow Q(x)) \\ &P(a) \\ &\therefore Q(a) \end{aligned}$$

Example.

Let

$$P(x) : x \text{ is divisible by 4}, \quad Q(x) : x \text{ is even.}$$

$$\begin{aligned} &\forall x \in \mathbb{Z}, (P(x) \rightarrow Q(x)). \\ &P(12). \\ &\therefore Q(12). \end{aligned}$$

That is:

$$\begin{aligned} &\text{Every integer divisible by 4 is even.} \\ &12 \text{ is divisible by 4.} \\ &\therefore 12 \text{ is even.} \end{aligned}$$

2. Universal Modus Tollens

$$\begin{aligned} &\forall x (P(x) \rightarrow Q(x)) \\ &\neg Q(a) \\ &\therefore \neg P(a) \end{aligned}$$

Example.

Let

$$P(x) : x \text{ is divisible by 4}, \quad Q(x) : x \text{ is even.}$$

$$\begin{aligned} &\forall x \in \mathbb{Z}, (P(x) \rightarrow Q(x)). \\ &\neg Q(7). \\ &\therefore \neg P(7). \end{aligned}$$

That is:

$$\begin{aligned} &\text{Every integer divisible by 4 is even.} \\ &7 \text{ is not even.} \\ &\therefore 7 \text{ is not divisible by 4.} \end{aligned}$$

3. Existential Modus Ponens

$$\begin{aligned} &\exists x P(x) \\ &\forall x (P(x) \rightarrow Q(x)) \\ &\therefore \exists x Q(x) \end{aligned}$$

Example.

Let

$$P(x) : x \text{ is divisible by 4}, \quad Q(x) : x \text{ is even.}$$

$$\begin{aligned} &\exists x \in \mathbb{Z}, P(x). \\ &\forall x \in \mathbb{Z}, (P(x) \rightarrow Q(x)). \\ &\therefore \exists x \in \mathbb{Z}, Q(x). \end{aligned}$$

That is:

There exists an integer divisible by 4.
Every integer divisible by 4 is even.
 $\therefore$ There exists an even integer.

D. Syllogisms with Quantifiers

1. Universal Syllogism

$$\begin{aligned} &\forall x (P(x) \rightarrow Q(x)) \\ &\forall x (Q(x) \rightarrow R(x)) \\ &\therefore \forall x (P(x) \rightarrow R(x)) \end{aligned}$$

Example.

Let

$P(x) : x$ is divisible by 6, $Q(x) : x$ is divisible by 3, $R(x) : x^2$ is divisible by 3.

Every integer divisible by 6 is divisible by 3.
Every integer divisible by 3 has square divisible by 3.
 $\therefore$ Every integer divisible by 6 has square divisible by 3.

2. Existential Syllogism

$$\begin{aligned} &\exists x P(x) \\ &\forall x (P(x) \rightarrow Q(x)) \\ &\therefore \exists x Q(x) \end{aligned}$$

Example.

There exists a prime number greater than 2.
Every prime number greater than 2 is odd.
 $\therefore$ There exists an odd number.

3. Universal-Existential Argument

$$\begin{aligned} &\forall x (P(x) \rightarrow Q(x)) \\ &\exists x P(x) \\ &\therefore \exists x Q(x) \end{aligned}$$

Example.

Every square number is nonnegative.
There exists a square number.
 $\therefore$ There exists a nonnegative number.

E. Equality Rules

1. Reflexivity of Equality

$$\therefore a = a$$

Example.

$$\therefore 7 = 7.$$

2. Symmetry of Equality

$$\begin{aligned} a &= b \\ \therefore b &= a \end{aligned}$$

Example.

$$\begin{aligned} x + 1 &= 5. \\ \therefore 5 &= x + 1. \end{aligned}$$

3. Transitivity of Equality

$$\begin{aligned} a &= b \\ b &= c \\ \therefore a &= c \end{aligned}$$

Example.

$$\begin{aligned} x &= 3. \\ 3 &= y. \\ \therefore x &= y. \end{aligned}$$

4. Substitution of Equals

$$\begin{aligned} a &= b \\ P(a) \\ \therefore P(b) \end{aligned}$$

Example.

Let

$$P(t) : t^2 = 9.$$

$$\begin{aligned} x &= 3. \\ P(3). \\ \therefore P(x). \end{aligned}$$

That is:

$$\begin{aligned} x &= 3. \\ 3^2 &= 9. \\ \therefore x^2 &= 9. \end{aligned}$$

F. Quantifier Negation Rules with Examples

1.

$$\neg(\forall x P(x)) \equiv \exists x \neg P(x)$$

Example.

$$\neg(\forall x \in \mathbb{R}, x > 0) \equiv \exists x \in \mathbb{R}, x \leq 0.$$

2.

$$\neg(\exists x P(x)) \equiv \forall x \neg P(x)$$

Example.

$$\neg(\exists x \in \mathbb{Z}, x^2 = 2) \equiv \forall x \in \mathbb{Z}, x^2 \neq 2.$$

3.

$$\neg(\forall x \exists y P(x, y)) \equiv \exists x \forall y \neg P(x, y)$$

Example.

$$\neg(\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, y > x)$$

is equivalent to

$$\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, y \leq x.$$

4.

$$\neg(\exists x \forall y P(x, y)) \equiv \forall x \exists y \neg P(x, y)$$

Example.

$$\neg(\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, x + y = y)$$

is equivalent to

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, x + y \neq y.$$

G. Invalid Forms to Warn Students About

1. Affirming the Consequent

$$\begin{array}{l} p \rightarrow q \\ q \\ \therefore p \end{array}$$

This is invalid.

Invalid Example 1.

If n is divisible by 4, then n is even.
 n is even.
 $\therefore n$ is divisible by 4.

This is invalid because 6 is even but not divisible by 4.

Invalid Example 2.

If it rains, then the ground is wet.
The ground is wet.
 $\therefore$ It rained.

This is invalid. The ground could be wet for another reason.

2. Denying the Antecedent

$$\begin{array}{l} p \rightarrow q \\ \neg p \\ \therefore \neg q \end{array}$$

This is invalid.

Invalid Example 1.

If n is divisible by 4, then n is even.
 n is not divisible by 4.
 $\therefore n$ is not even.

This is invalid because 6 is not divisible by 4, but 6 is even.

Invalid Example 2.

If I study, then I pass.
I did not study.
 $\therefore$ I did not pass.

A person may pass for another reason.

3. Invalid Quantifier Reversal

$$\forall x \exists y P(x, y) \not\equiv \exists y \forall x P(x, y)$$

Invalid Example.

The statement

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, y > x$$

is true, since we may choose $y = x + 1$.

But

$$\exists y \in \mathbb{R}, \forall x \in \mathbb{R}, y > x$$

is false, because no single real number is greater than every real number.

4. Invalid Generalization from One Example

$$\begin{array}{l} P(a) \\ \therefore \forall x P(x) \end{array}$$

Invalid Example.

2 is prime.
 $\therefore$ Every natural number is prime.

Activity : Valid or Invalid?

For each argument:

- Translate into symbolic form
- Identify the argument pattern
- Decide whether it is valid or invalid

1. If n is prime and greater than 2, then n is odd. n is prime and greater than 2. Therefore, n is odd.
2. If a student attends lectures, then the student understands the course. The student understands the course. Therefore, the student attended lectures.
3. If x is rational, then x^2 is rational. x^2 is not rational. Therefore, x is not rational.
4. Either the relation is reflexive or it is not reflexive. The relation is not reflexive. Therefore, the relation is reflexive.
5. If a function has an inverse, then it is one-to-one. This function is not one-to-one. Therefore, this function does not have an inverse.

Part 4: Testing Validity Using Truth Tables

To test whether an argument is valid, form the implication:

$$(\text{Premise 1} \wedge \text{Premise 2} \wedge \cdots) \rightarrow \text{Conclusion}.$$

If this implication is a tautology, the argument is valid.

Example

Consider:

$$\begin{array}{l} p \rightarrow q \\ p \\ \therefore q \end{array}$$

The associated statement is:

$$((p \rightarrow q) \wedge p) \rightarrow q.$$

If this is a tautology, then the argument is valid.

Note

One does not need to use truth tables for every argument, but truth tables help justify validity formally.

Reflection and Exit

Reflection

- Can an argument be valid even if its conclusion is false?
- Can an argument be invalid even if its conclusion is true?
- Why is validity important in mathematical proof?

Common Misconceptions

Misconception	Correction
A valid argument must have true premises	Validity is about logical form
True conclusion means valid argument	An invalid argument can have true conclusion
$p \rightarrow q$ means $q \rightarrow p$	The direction matters
Affirming the consequent is valid	It is invalid
Denying the antecedent is valid	It is invalid

Independent Study

Practice

For each argument:

- Identify premises and conclusion
 - Translate into symbols
 - Decide whether it is valid
1. If n is divisible by 6, then n is divisible by 3. n is divisible by 6. Therefore, n is divisible by 3.
 2. If $x > 2$, then $x^2 > 4$. $x^2 > 4$. Therefore, $x > 2$.
 3. If a relation is an equivalence relation, then it is reflexive. The relation is not reflexive. Therefore, it is not an equivalence relation.
 4. Either n is even or n is odd. n is not odd. Therefore, n is even.

Investigation Task

Find two examples of arguments from everyday life. For each example, identify whether the reasoning is valid or invalid.

Reflection Notes

Observe whether you:

- Confuse truth with validity
- Reverse the direction of implications
- Misidentify premises and conclusions
- Recognize invalid argument forms

Try to:

- Write arguments symbolically
- Read symbolic arguments with proper understanding
- Check whether true premises force the conclusion
- Compare similar-looking valid and invalid forms